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What to benchmark?
A systematic approach and cases

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Abstract *This paper presents an approach for systematically defining the object of study of benchmarking based on deriving improvement actions from customer expectations and strategic decisions through business processes, and prioritising improvement actions that will most contribute to strategic objectives. It is based on management concepts such as business process mapping, performance measurement and other tools. After some introductory theoretical background and discussion of the proposal, a set of steps to guide the implementation of such an approach is presented and detailed. Finally, two cases of application are presented to illustrate the validity as well as to make some considerations based on the potential benefits and limitations of this proposal.*

Introduction

In an era when many organisations are trying to catch up with global competition, rapid technological advances and ever-changing customer needs, benchmarking has been gaining attention among managers and academics as a means of strengthening a company's ability to compete. Benchmarking has been defined as the search for industry best practices that will lead to superior performance (Camp, 1989). This definition has been coined by Robert Camp, who first wrote a book on the subject based on his experience at Xerox Corporation in the USA.

In a more elaborated way, the International Benchmarking Clearing House Design Committee defines benchmarking as "... a systematic and continuous measurement process; a process of continuously measuring and comparing an organisation's business process against business leaders anywhere in the world to gain information which will help the organisation to take action to improve its performance" (Lema and Price, 1995).

Although benchmarking in business organisations is a relatively new concept and practice, it has rapidly gained acceptance worldwide as an instrument of continuous improvement in the context of total quality management (TQM). In the USA, where it was first introduced, a large number of organisations make use of benchmarking. Many are the organisations in the USA and Europe that promote the use of benchmarking, such as the International Benchmarking Clearing House or the The European Network for Advanced Performance System (ENAPS), which provide benchmarking databases and assistance in identifying partners. Following this international



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trend, many organisations in Brazil, from large to medium size, are adopting benchmarking.

The activity of benchmarking can be decomposed into basically five steps (Drew, 1997):

(1) identify the object of study;

(2) select the superior performer (benchmarking partner);

(3) collect and analyse data;

(4) set performance goals for improvement; and

(5) implement plans and monitor results.

One can see there is a clear relationship between benchmarking and improvement strategies in the sense that if it is to be used as part of the business strategy for gaining and maintaining competitive advantage, improvement projects must be prioritised by taking into consideration the company performance levels relative to competitors on product and operational aspects most important to present and future market demands. In fact, most models and methodologies for benchmarking implementation published in the literature (Zairi and Leonard, 1994) stress the importance of aligning benchmarking projects with competitive and operations strategy, so that organisational efforts for improvements are directed towards dimensions of performance critical to competitiveness.

However, many companies, in their attempts to adopt world class management practices rapidly such as benchmarking, tend to adopt a strong operational view of improvement, devoting little or no attention at all to the alignment of such practices with market demands and strategic objectives. This is especially true for medium to small size companies, where strategic thinking and positioning is part of the tacit knowledge of the organisation.

Taking into account these initial considerations, this article proposes determining what to benchmark by systematically deriving improvement actions from customer expectations and strategic decisions through business processes, and prioritising benchmarking projects on subjects that will most contribute to strategic objectives. In the following sections, a brief introduction is made on the subject of benchmarking so as to introduce the proposed approach and application cases.

Benchmarking types, methodologies and services

Benchmarking practices can be generically classified according to the nature of the object of study of benchmarking and the partners against whom comparisons are made. In terms of object of study, benchmarking can be classified as:

- process benchmarking: used to compare operations, work practices and business processes;
- product benchmarking: used to compare products and/or services;

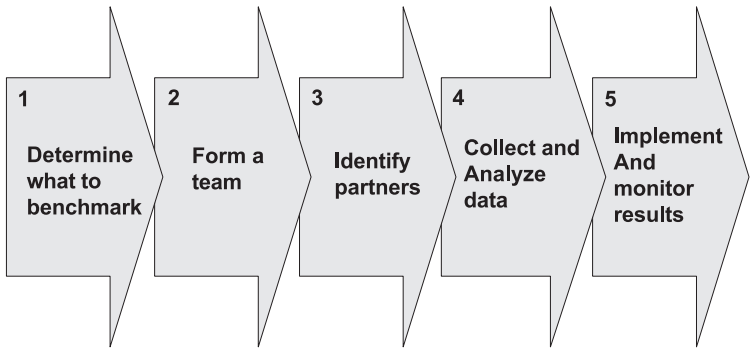
- strategic benchmarking: used to compare organisational structures, management practices and business strategies. In a sense, it posses some similarities to process benchmarking.

Benchmarking partners may include other units of the same organisation, competitors in the same or different geographical markets and organisations in related or unrelated industries, in the same or different countries.

Classifications of benchmarking found in the literature are mainly based on the type of partner, as follows (Camp, 1989):

- *Internal benchmarking*. By comparison of performance of units or departments within one organisation. Even though not explicit in this definition, comparison can also be made of similar products or services of similar business units.
- *Competitive benchmarking*. By comparison of performance with direct product competitor. In this case, comparison can be made of products or services and business processes. Reverse engineering is a term more appropriate for product benchmarking.
- *Functional benchmarking*. Specific function comparison with best practice. It is an application of process benchmarking that compares a particular business function in two or more organisations in the same industry.
- *Generic benchmarking*. Search for the best practice irrespective of industry. It is similar to functional benchmarking but the aim is to compare with the best in class without regard to industry.

The most publicised benchmarking procedure is due to Camp (1989) and was originally developed and applied at Xerox Corporation as part of its TQM effort. Other models are described by Zairi and Leonard (1994). Although different companies have adopted different models to benchmarking, most of these models generally conform to a process comprising five basic steps, as shown in Figure 1 (Drew, 1997).



Source: Drew (1997)

Figure 1.
Generic benchmarking process

Several organisations offer benchmarking services that facilitate the development of benchmarking projects. For instance, The Benchmarking Exchange© enables members, among other things, electronically to locate and communicate with potential benchmarking partners, to research best in class companies, find out how they achieved best in class performance or to seek assistance and advice from others who have already benchmarked what a company is about to benchmark. Benchmarking services are also provided by the Benchmarking Clearinghouse of the American Productivity and Quality Centre (AQPC).

Well-publicized examples of benchmarking are the cases of Xerox Corporation, LL Bean (Camp, 1989), Alcoa, Motorola, AT&T, Florida Power and Light and General Electric (Longbottom, 2000).

Reviewing the literature, it is possible to find many examples of applications of benchmarking, ranging from manufacturing to non-manufacturing organizations including education (Moreland *et al.*, 2000), public sector administration (Kouzmin *et al.*, 1999), health care (Bullivant, 1996) and many others. In the manufacturing industry, benchmarking focuses on subjects such as product innovation (Dacko, 2000), logistics (Bagchi, 1997), maintenance (Muthu *et al.*, 2000), human resource management (Rodwell *et al.*, 2000) and quality assurance among others.

In a survey conducted by Voss *et al.* (1997) involving a sample of over 600 European manufacturing companies, it is shown that increased levels of benchmarking use were associated with higher levels of both adoption of best practices and operational performance. Based on the study, the authors proposed a relationship model between learning, benchmarking, understanding and performance, in which benchmarking, as part of a learning process of an organization, at the same time promotes a company's understanding of its strengths and weaknesses and higher levels of performance.

The practice of benchmarking has also been characterized by Longbottom (2000) in a survey involving over 200 British organizations. In his study he reported, among other things, that very little evidence was found to show that organizations are identifying and prioritising projects based on their corporate and strategic planning process. According to his study, selection of projects is largely *ad hoc*, mostly arising from project/benchmarking champions, based on the need to update equipment or technology, or reacting to rising costs/falling profits. This lack of alignment between strategic planning and benchmarking projects, considered by the author as an important issue for future research, is the central point of this article, as better discussed hereafter.

Defining benchmarking subjects

Defining the object of study of a benchmarking initiative is a first and fundamental step if improvements on product or operational performance intend to boost competitiveness and business results. Therefore, the selection of a product or process for benchmarking must be preceded by a diagnostic of the

current situation and an analysis of factors of success or top priority dimensions of product and process performance (which in turn depend upon strategic decisions about what market segments and dimensions it wishes to compete on) to deliver expected value to customers. Also, since customer expectations are also influenced by what competition can offer, analysis of performance against competitors should be considered in order to decide what product or process should be prioritised for benchmarking.

Most of the benchmarking literature is frequently more concerned with how to develop a benchmarking project once the object of study has been defined. Focused on the question of determining what to benchmark are the studies presented by Partovi (1994) and Büyüközkan and Maire (1998). Partovi (1994) proposes the use of the analytic hierarchy process (AHP) as a means to prioritise benchmarking projects. Büyüközkan and Maire (1998) propose the use of principal component analysis (PCA) and common factor analysis (CFA) also as prioritisation tools. In both cases, the tools are not of simple use, requiring special software to handle the calculations. Moreover, using a quantitative method of analysis is no guarantee that a proper course of analysis of qualitative data has been performed. Therefore this paper is more concerned with the steps along the path of deploying product and process improvement needs into benchmarking projects rather than focusing on a particular tool for decision making.

Although benchmarking procedures starts by defining what should be benchmarked, there is generally no clear procedure for identifying improvement needs. One reason for that is the fact that benchmarking projects are often initiated by managers who, having a strong operational view of improvement, assume process redesign and improvement can be made regardless of business and manufacturing strategy. Still another reason is the functional organization of most companies (as opposed to a process based organization), which can leave the process of defining what should be benchmarked subject to functional boundaries and constraints.

Therefore, aligning benchmarking projects with company-wide improvement needs demands an analysis in which projects selected for implementation are those that can most contribute to the efficiency and effectiveness of business process mostly related to prioritised competitive criteria. That is (Figure 2):

- competitive dimensions are prioritised for improvement based on strategic positioning, customer expectations and performance on such dimensions against competition;
- priority processes for improvement are selected based on the impact of such processes on dimensions taken as improvement priorities as well as on qualitative and/or quantitative diagnostics of performance;
- benchmarking projects are defined based on the extent of their contribution to levering product or process performance.

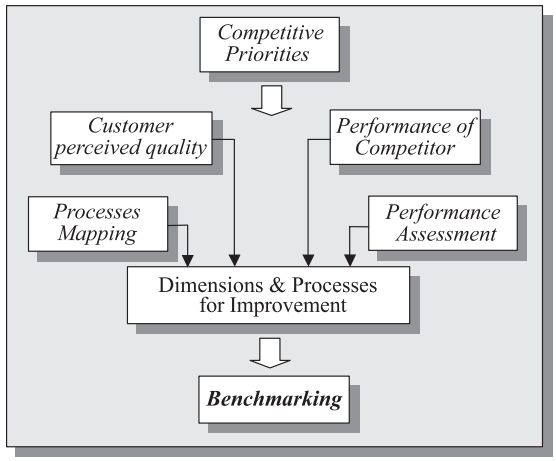


Figure 2.
Elements in defining what to benchmark

Thus, the approach presented in this paper is very much concerned with identifying dimensions of performance of product and operations, as well as mapping business processes involved in delivering value to customers. Competitive dimensions such as quality, cost, delivery and flexibility can be disaggregated into more focused aspects such as conformity, reliability, speed, dependability, cost of maintenance, flexibility of process, etc. Other more tacit differentiating aspects can be gathered through researching target markets.

Moreover, since operations are in general managed from a functional perspective, mapping business processes is a fundamental step to understanding the flow of information and resources through the business processes of the internal value chain. For instance, the AQPC Benchmarking Clearing House (APQC, 2000) defines 11 macro-processes, including support processes. Simplified flowcharting tools or information technology tools for business process representation, such as ARIS© (architecture for integrated information system) or Visio©, can be used for this purpose.

Mapping can also bring the benefit of helping in assessing performance of operational and supporting processes. Assessment of performance is essential to diagnose the root causes of problems or weakness so as to determine what to benchmark. Information on process performance can be gathered with qualitative assessment of the current situation. Cause and effect diagrams such as the fish-bone diagram may be used for this purpose. The current reality tree (Dettmer, 1997) from the theory of constraints may also be used. It works by identifying cause-and-effect interrelationships among potential root causes and undesirables effects or problems. Apart from qualitative analysis of performance, quantification of performance on critical dimensions can also be used as one of the instruments for guiding the identification of processes and dimensions of operations for benchmarking.

Steps for defining the object of study of benchmarking

The considerations made so far suggest that the process of defining what to benchmark may be organised into a sequence of steps as proposed in Figure 3.

Step 1: product and market analysis

Gather information on product characteristics, target customer and markets, competitive priorities, manufacturing and financial strategies and general areas for improvements. This will help to understand what dimensions and activities are most crucial to competitiveness.

Step 2: critical dimensions

Gather information on customer expectations and perceived quality for different categories of customers/products and rank relative importance of requisites for most important customers; also, gather information on performance against competitors in attending customer expectations. This helps identify dimensions most in need of improvement.

Step 3: critical processes

Map all the processes and activities belonging to or supporting the value adding chain and understand their relationship with the dimensions most in need of improvement. This may be helped by constructing a matrix relating processes to dimensions. This will help to focus on the attention on the

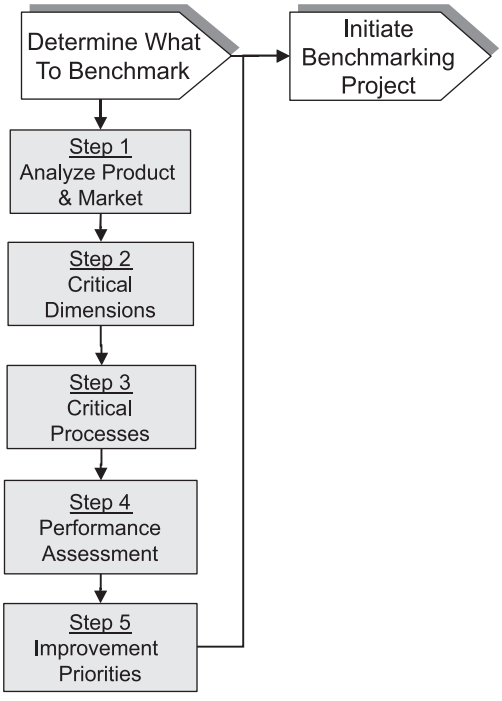


Figure 3.
Steps for defining the object of study of benchmarking

processes and activities that most impact performance on prioritised competitive dimensions.

Step 4: performance assessment
Conduct a qualitative or quantitative assessment of performance of the critical processes and activities. A diagnosis of current situation is of fundamental importance to realising what areas or activities are the weak points and need to be addressed. Quantitative information, if available, can reveal areas and dimensions in need of improvement.

Step 5: improvement priorities
After performing the analysis proposed in steps 1 to 5, the dimensions and activities most in need of improvements become evident. From this point onwards, the benchmarking project itself can start for those subjects for which a benchmarking application is considered to be adequate.

Research method

The sort of research reported in this paper essentially proposes a solution and test through action whether the conceived solution brings the expected benefits. It is therefore part of a larger research cycle in which theory development guides action, which in turn helps further theory adjustments.

Thus, after conceptual formalization, the next step was concerned with developing cases of application of the proposed steps so as to be able to test and discuss the approach described in the previous section. For that, two companies were selected based on their interest in applying benchmarking for improvement action.

In both cases, a team comprising researchers and people from the companies was set up so as to make sure that the decision process taken implicit in the proposal application, as well as priority definitions, would be the result of full interaction between researchers and the company's staff. The cases are described in the next sections.

First application case

Company description
A first application of the steps proposed above was performed in a middle-sized company located in the State of São Paulo, Brazil. It develops, manufactures and markets residential steel doors and windows, with a daily production volume of about 8,000 pieces, comprising nearly 600 different catalogue items. It works on an assembly to order scheme. Since it has been experiencing increasing competition in its market segments it has welcomed our proposal of following the approach described above so as to identify improvement needs to be tackled by a benchmarking project. An improvement team was formed comprising members of company staff and a graduate student, to guide the deployment process.

Step 1: product and market analysis

The company works on basically three different product lines, focused on three different market segments. In the segment of low cost products, competition is based on cost, although the quality of manufacturing conformance is also of importance. In the segment of styled products, although cost is still a competitive priority, product quality, including design, durability and robustness, is in general a winner criterion. For these two market segments, the first layer clients are the construction stores. For the shop owners, the speed and reliability of delivery are also important in their buying decision. Finally, the third segment comprises the construction industry of government developed residential suburbs. For this segment, quality of product, especially durability and robustness, cost and on-time delivery are competitive criteria.

Steps 2 and 3: critical dimensions and processes

After analysis of products and market segments, it became clear that critical competitive dimensions for improvement were product quality, quality of conformity and delivery. Having defined the critical dimension, the next step was to focus on the critical processes and activities. The value adding chain of the company is made up of basically three macro-processes:

(1) product and process development;

(2) marketing and selling; and

(3) production and delivery.

Analysis of the relationship among competitive dimensions and business processes has shown the importance of concentrating efforts on the processes of product and process development and production and delivery.

Step 4: performance assessment

To assess company performance on such processes the team has conducted a qualitative assessment through the elaboration of a current reality tree. As a result of this process, the team could realize that the main problems were related to the quality of conformity and delivery time. The team also realized that the root causes of poor performance were basically related to the lack of an adequate production planning and control system, and equipment set-up activities not adequately performed.

Step 5: improvement priorities and benchmarking project

To address these problems, the team proposed improvement actions on three fronts: production planning and control system; methodology to analyse and solve product and process quality problems; and set-up activities. To tackle the improvement activity related to equipment set-up, the team understood that searching for good practices could lead to good results. A benchmarking project was then initiated by defining a team and partners. By contacting the equipment manufacturer, it was possible to get in contact with other equipment users and establish conversation to initiate a benchmarking project. After four

months of work the team could implement good practices of machine set-up that not only reduced set-up times but also improved quality of conformance, helping the company to improve its competitiveness to the point of recently receiving an award from an industry association as the top company in its segment.

Second application case

Company description

A second case of application was developed in a small company located in the State of São Paulo that develops, manufactures and commercialises a full range of equipment for tuning electronic fuel injection. It employs about 160 people and has an annual income of around US$2 million. Although it is the only company that manufactures such products in Brazil, it has been facing competition from abroad. Therefore the owners of the company welcomed our proposal to identify improvement needs to be tackled by a benchmarking project. An improvement team was formed, comprising in this case one member of the company staff (sales manager) and a graduate student, to guide the deployment process.

Step 1: product and market analysis

For the case under consideration, the team focused on the main product of the range of equipment for tuning electronic fuel injection manufactured by the company. This product presents several attributes that differentiate it in its market segment. Its clients are garage owners all over the country that are still not prepared for using high-tech equipment. Therefore, after-sales technical training and assistance are of fundamental importance to keep and increase participation in the market. Also, the company is the only one to manufacture this product in Brazil, and therefore it retains most of the Brazilian market. However, the company has to be prepared to compete abroad as well as face competition in the in the local market.

Steps 2 and 3: critical dimensions and processes

To identify critical dimensions, the team decided to get in touch with its clients so as to get a non-biased understanding of requirements of products and services. In total, a sample of 30 of its clients (basically garages all over the country) was interviewed. From the interviews, it was confirmed that responsiveness and completeness of after-sales services were the most important dimensions to keep competitive advantage. This fact has shown the importance of the activities of the processes of marketing and selling, and after-sales service.

Step 4: performance assessment

Through qualitative assessment of these processes the team could also realize that the main problems were related to lack of technical knowledge of sales people and inadequate after-sales technical assistance. Therefore the root causes of poor performance were basically related to training of sales and technical assistance people.

Step 5: improvement priorities and benchmarking projects

From this point, the team understood that searching for good practices regarding after-sales training and technical assistance could lead to good results. A benchmarking project was then initiated by, after defining a team, searching for partners in the medical and automation equipment sectors. At the time of reporting this case, the benchmarking project was still waiting to be started, with no results to be reported.

Final considerations

This paper has emphasised the importance of benchmarking practice as a means to promote continuous improvement in organisational performance. Also, it has stressed the importance of initiating a benchmarking project only after going through a detailed analysis of competitive dimensions and business processes most in need of improvement. In doing that the benchmarking team not only guarantees that the project will be aligned with strategic objectives and needs, it also provides a basis for learning what a company's weakness and strengths are.

The cases presented have shown to be good examples of how a company can apply the proposed set of steps for deploying benchmarking initiatives focused on processes critical to competitive dimensions. Although very simple, the cases illustrate the importance of: deploying benchmarking from strategic decisions, customer expectations and performance in the face of competition; deploying improvement needs through the business processes of the organisation and not through the functional hierarchy; and prioritising processes for improvement that can most positively impact performance on dimensions critical to competitiveness.

The application of the proposed steps have shown that priority definition of competitive dimensions or improvement initiatives and deployment of activities from competitive dimensions or benchmarking projects from assessment, if not properly performed, may lead to inconsistent results. Therefore further work has to be done to incorporate in the method tools that help to guide the process of deployment and priority definition, as well as a procedure for systematically revising the decisions taken to insure consistency all over the process.

At first it may appear that managers already know what has to be improved in operations to remain or gain competitiveness, dispensing with a more elaborate approach. However, as companies get better operations performance, improvement needs and actions become far less evident and therefore they may need to rely on a more systematic and step-by-step approach. However, it should be noted that this approach is particularly suitable for companies that rely mainly on operational effectiveness for gaining advantage over competition, as in the cases presented, or manufacturers of components for consumer durable products. For companies with strategies based mainly on innovation of products or services, although performing benchmarking is equally important, operations improvement will probably not be priority.

Finally, it should also be noted that although benchmarking practice is central to the approach for improvement deployment presented in this paper, it encompasses different practices such as process modelling, performance assessment and other tools and techniques. Therefore, further research has to be done dealing with integrating such practices under a framework of management of improvement and change, including managing feedback of information on progress review so as to guarantee the continuity of the improvement and changing process.

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